17. The pH of 10^{-5}M KOH solution will be

a) 9 b) 5 c) 19 d) none of these

18. $H_2PO_4^-$ the conjugate base of

a) PO_4^{3-} b) P_2O_5 c) H_3PO_4 d) HPO_4^{2-}

19. Which of the following can act as Lowry – Bronsted acid as well as base?

a) HCl b) SO_4^{2-} c) HPO_4^{2-} d) Br^-

20. The pH of an aqueous solution is Zero. The solution is

a) slightly acidic b) strongly acidic c) neutral d) basic

21. The hydrogen ion concentration of a buffer solution consisting of a weak acid and its salts is given by

a) $[H^+]=\frac{K_a[acid]}{[salt]}$ b) $[H^+]= K_a[salt]$ c) $[H^+]=K_a[acid]$ d) $[H^+]=\frac{K_a[salt]}{[acid]}$

22. Which of the following relation is correct for degree of hydrolysis of ammonium acetate?

a) $h = \sqrt{\frac{K_h}{C}}$ b) $h = \sqrt{\frac{K_a}{K_b}}$ c) $h = \sqrt{\frac{K_w}{K_a.K_b}}$ d) $h = \sqrt{\frac{K_a.K_b}{K_w}}$

23. Dissociation constant of NH_4OH is 1.8×10^{-5} the hydrolysis constant of NH_4Cl would be

a) 1.8×10^{-19} b) 5.55×10^{-10} c) 5.55×10^{-5} d) 1.80×10^{-5}

Answer the following questions:

1. What are Lewis acids and bases? Give two example for each.
2. Discuss the Lowry – Bronsted concept of acids and bases.
3. Indentify the conjugate acid base pair for the following reaction in aqueous solution

i) $HS^-(aq) + HF \rightleftharpoons F^-(aq) + H_2S(aq)$ ii) $HPO_4^{2-} + SO_3^{2-} \rightleftharpoons PO_4^{3-} + HSO_3^-$

iii) $NH_4^+ + CO_3^{2-} \rightleftharpoons NH_3 + HCO_3^-$

4. Account for the acidic nature of $HClO_4$ in terms of Bronsted – Lowry theory, identify its conjugate base.
5. When aqueous ammonia is added to $CuSO_4$ solution, the solution turns deep blue due to the formation of tetramminecopper (II) complex, $[Cu(H_2O)_4]^{2+}_{(aq)}+4NH_3(aq) \rightleftharpoons [Cu(NH_3)_4]^{2+}_{(aq)}$, among H_2O and NH_3 Which is stronger Lewis base.
6. The concentration of hydroxide ion in a water sample is found to be 2.5×10^{-6}M. Identify the nature of the solution.

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